

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T28 · Forming and Solving Equations

With Answers

Questions with full worked solutions

■ 7 pages

FAMILY MAP

Forming Solving Equations

F01

Form and solve a quadratic from data

2 Q | 9 marks

F02

Derive a general expression or formula

1 Q | 3 marks

- 5 Yasmin has some identical rectangular tiles.
Each tile is L cm by W cm.

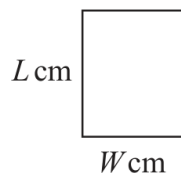


Diagram **NOT**
accurately drawn

Using 9 of her tiles, Yasmin makes rectangle $ABCD$, shown in the diagram below.

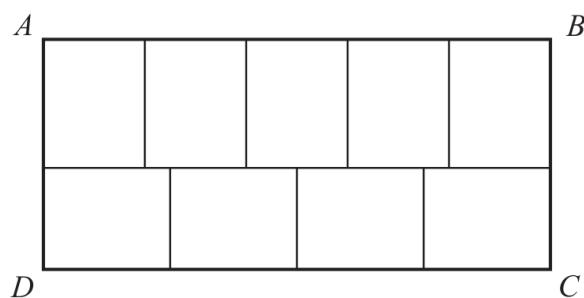


Diagram **NOT**
accurately drawn

The area of $ABCD$ is 1620 cm^2

Work out the value of L and the value of W .

$L = \dots\dots\dots$ $W = \dots\dots\dots$

Worked solution

November 2021 | Question 5 | 5 marks

Family: Form and solve a quadratic from data | Variant: Area or dimension quadratic model

METHOD / WORKING

1. The top row has 5 tile widths and the bottom row has 4 tile lengths, so

2. $5W=4L$

3. $L=\frac{5}{4}W$

4. The rectangle has width $5W$ and height $L + W$:

5. $5W(L+W)=1620$

6. Substitute $L = \frac{5}{4}W$:

7. $5W\left(\frac{5}{4}W+W\right)=1620$

8. $5W\left(\frac{9}{4}W\right)=1620$

9. $\frac{45}{4}W^2 = 1620$

10. $W^2 = 144$

11. $W=12, L=\frac{5}{4}(12)=15$

12. Answer: $L = 15\text{cm}, W = 12\text{cm}$.

FINAL ANSWER

L=15 cm, W=12 cm.

- 2 Here are two rectangles, rectangle A and rectangle B .

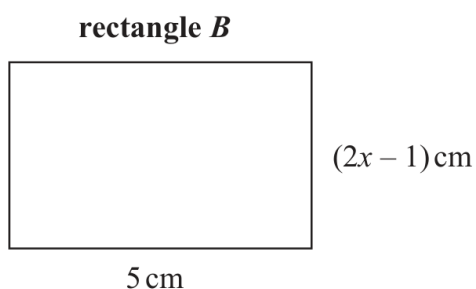
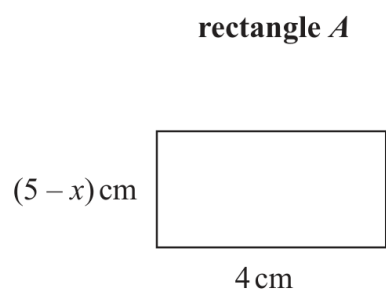


Diagram **NOT**
accurately drawn

The area of rectangle B is twice the area of rectangle A .

Work out the value of x .

Show your working clearly.

$x = \dots\dots\dots$

Worked solution

November 2021 | Question 2 | 4 marks

Family: Form and solve a quadratic from data | Variant: Area or dimension quadratic model

METHOD / WORKING

1. Area of rectangle A:

2. $4(5-x)$

3. Area of rectangle B:

4. $5(2x-1)$

5. Rectangle B has twice the area of rectangle A:

6. $5(2x - 1) = 2 \cdot 4(5 - x)$

7. $10x-5=40-8x$

8. $18x=45$

9. $x=2.5$

10. Answer: $x = 2.5$.

FINAL ANSWER

$x=2.5.$

20 Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by $x = \frac{-4 \pm 2\sqrt{39}}{10}$

Find the value of a and the value of c

Show your working clearly.

$a =$

$c =$

(Total for Question 20 is 3 marks)

Worked solution

November 2025 | Question 20 | 3 marks

Family: Derive a general expression or formula | Variant: General geometric formula derivation

METHOD / WORKING

Worked solution
November 2025 Question 20 3 marks
Family: Derive a general expression or formula Variant: General geometric formula derivation
Method
For
$ax^2 + 4x + c = 0$
the quadratic formula gives
$x = \frac{-4 \pm \sqrt{4^2 - 4ac}}{2a}$
The given solutions are
$x = \frac{-4 \pm 2\sqrt{39}}{10}$
Compare denominators:
$2a = 10$
$a = 5$
Compare the square root parts:
$\sqrt{16 - 4ac} = 2\sqrt{39}$
Square both sides:
$16 - 4ac = 156$
Use $a = 5$:
$16 - 20c = 156$
$-20c = 140$
$c = -7$
Answer: $a = 5, c = -7$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER